

Methods of Separation in Everyday Life

Class 6 Science · Chapter 9 · Study Notes

1. Why Do We Separate Substances?

Most of the things we use are mixtures of different substances. Rice from the field has stones and husk mixed with it. Muddy water has soil mixed in it. Sea water has salt dissolved in it. We separate the useful part from the unwanted part, or separate the components to use each one. The method we choose depends on the kind of mixture.

2. Handpicking

Handpicking is used when the pieces to be separated are large enough to be picked out by hand. For example, we pick out small stones, insects and damaged grains from wheat or pulses before cooking. It is a simple method, but it works only for small quantities.

3. Threshing

Threshing is the process of separating grains from the stalks of harvested crops. The stalks are beaten, or bullocks or machines are used to make the grains fall off. Farmers thresh wheat and paddy after harvesting.

4. Winnowing

Winnowing is used to separate lighter husk from heavier grains with the help of wind. The mixture is dropped from a height. The heavier grains fall in a heap near the person, and the lighter husk is blown away to a distance by the wind. Farmers use winnowing after threshing. A machine called a winnower can be used instead when there is not enough wind.

5. Sieving

Sieving separates particles of different sizes. A sieve has tiny holes. Smaller particles pass through the holes, and larger ones stay on top. Flour is sieved to remove the bran and lumps. At construction sites, sand is sieved to remove pebbles.

6. Sedimentation and Decantation

When muddy water is left standing, the heavier soil particles slowly settle at the bottom. This settling is called sedimentation. The clear water above can then be poured gently into another container without disturbing the sediment. This pouring off is called decantation.

Sedimentation and decantation are used when the solid is heavier than the liquid and does not dissolve in it. For example, they are used to separate water and sand, or to separate pulses that are washed in water.

7. Filtration

Filtration is used to separate insoluble solids from a liquid using a filter, such as filter paper or a piece of cloth. The liquid passes through the tiny pores of the filter, and the solid stays behind on it. The liquid that comes out is called the filtrate. Tea leaves are separated from tea with a tea strainer, which is a kind of filter. Water purifiers use filters to remove dirt.

Filtration can be used to make muddy water clearer, but the water may still contain harmful germs, so it must be boiled or treated before drinking.

8. Evaporation

Evaporation is used to separate a dissolved solid from a liquid. When salt water is heated, the water turns into vapour and goes into the air, and the salt is left behind. In coastal areas, salt is obtained from sea water by allowing the water to evaporate in shallow pans in the heat of the Sun.

9. Magnetic Separation

A magnet can separate magnetic materials such as iron from a mixture with non-magnetic materials such as sand or sawdust. Iron pins mixed with sand are pulled out when a magnet is moved over the mixture. This method is used in factories to remove iron pieces from grain and from scrap.

10. Separating a Mixture of Sand, Salt and Water

To get sand, salt and clean water from a mixture, we can follow these steps in order.

- Filter the mixture. The sand stays on the filter paper, and the salt water passes through.
- Heat the salt water in a dish. The water evaporates, and the salt is left in the

dish.

- If clean water is needed as well, the vapour can be cooled so that it condenses back into water.

11. Quick Recap

- Handpicking, winnowing, sieving, sedimentation, decantation, filtration, evaporation and magnetic separation are common separation methods.
- Choose the method according to the properties of the substances, such as size, weight, solubility and magnetism.
- Winnowing uses wind. Sieving uses size. Filtration removes insoluble solids from a liquid. Evaporation recovers a dissolved solid.